BhumiSetu

India's AI-Powered Land Decision Platform

Document Type: Product Requirements Document (PRD)

Target Audience: Investors, Engineering, Product, Design

Version: 1.0.0 (Master Blueprint)

Confidentiality: Strictly Confidential

PART 1: Strategy & Vision

Executive Summary

BhumiSetu is the foundational operating system for India's land ecosystem. It resolves the intrinsic opacity of the Indian real estate and land sector by unifying AI-driven insights, structured workflow automation, verified professional networks, and geospatial intelligence. We do not sell listings; we sell confidence, clarity, and execution.

The Moat & Category Design

We are creating a new category: **Land Decision Infrastructure**. Our economic moat is derived from:

- **Land Decision Graph:** Connecting parcels, legal histories, spatial data, and transaction flows into a unified knowledge graph.
- **AI Learning Loop:** Every completed workflow, verified document, and resolved case trains our proprietary AI models, making subsequent decisions faster and more accurate.
- **Professional Network Effects:** As more verified surveyors and lawyers join to receive workflow, they bring their own clients to the platform to manage cases, driving organic, B2B2C acquisition.

AI Philosophy: AI does not replace human professionals. It organizes, analyzes, summarizes, and coordinates. Human experts remain accountable for field verification, legal binding, and ultimate advisory.

PART 2 & 3: Market Research & Personas

Indian Land Industry & Gap Analysis

The Indian land market is characterized by high friction, low trust, fragmented data, and asynchronous offline processes. Current solutions focus on the "Search" phase (listings like MagicBricks, 99acres) but completely abandon the user during the "Diligence & Execution" phase. BhumiSetu bridges this exact gap.

Primary Personas

1. The Anxious Buyer (Retail)

Pain Point: Fear of litigation, encroachments, and fake documents.

Goal: Absolute certainty before deploying life savings into a land parcel.

Platform Value: A transparent dashboard tracking legal diligence, site survey progress, and a "BhumiSetu Trust Score" for the parcel.

2. The Remote NRI Owner (Retail)

Pain Point: Inability to physically monitor land, fear of squatter encroachment.

Goal: Secure digital asset management and remote verification.

Platform Value: Subscription-based satellite monitoring alerts and on-demand physical audit workflows.

PART 4 & 5: Ecosystem & Information Architecture

The Product Ecosystem

- **Customer App (Web/Mobile):** The primary interface for buyers/sellers to initiate cases, track progress, make payments, and view reports.
- **Partner App (Mobile First):** For field surveyors and lawyers. Features offline GPS tagging, document uploads, task checklists, and earning tracking.
- **Enterprise Portal:** High-volume interface for Banks (loan diligence), Solar Co's (land acquisition), and Developers.
- **Bhumi Core (Admin/AI):** The backend workflow engine and LLM orchestration layer.

PART 12: AI Architecture

BhumiSetu relies on a Multi-Agent LLM Architecture to automate the heavy lifting of land diligence.

Document Intelligence Agent (DIA)

Function: Ingests unstructured, multi-lingual land records (7/12 extracts, Khataunis, Sale Deeds). Uses OCR and Vision-Language Models.

Tech: RAG pipeline with vector embeddings (Pinecone/Milvus), utilizing fine-tuned IndicLLMs for regional language translation.

Output: Structured JSON summarizing ownership history, encumbrances, and missing chain-of-title links.

Workflow Orchestrator Agent

Function: Evaluates the output from the DIA and dynamically constructs the required workflow.

Example: If DIA flags an unclear boundary, the Orchestrator automatically dispatches a task to a local Partner Surveyor via the Partner App.

PART 13: Database Design (Core Entities)

Entity	Attributes	Relationships
Cases	id, user_id, status, parcel_id, created_at, workflow_state	Belongs to User, Has Many Tasks, Has One Parcel
Parcels	id, geojson_polygon, state, district, survey_number, trust_score	Has Many Documents, Has Many Cases
Documents	id, case_id, type, s3_url, extraction_json, verification_status	Belongs to Case, Belongs to Parcel
Partners	id, type (lawyer, surveyor), rating, service_area, kyc_status	Has Many Tasks, Has Many Payouts

PART 14: API Design (Snippet)

RESTful API utilizing stateless JWT authentication. Rate limited via API Gateway.

```
POST /api/v1/cases/initiate
```

Payload: `{ "service_type": "legal_diligence", "parcel_data": { "state": "MH", "survey_no": "142/B" } }`

Response: `{ "case_id": "c_987xyz", "status": "pending_documents", "assigned_agent": "DIA_v2" }`

PART 15: Technology Stack

- **Frontend:** Next.js (React), Tailwind CSS, Framer Motion (for premium minimal animations).
- **Mobile Apps:** React Native (Expo) - unified codebase for iOS/Android.
- **Backend:** Node.js (NestJS) for API gateway and workflows. Python (FastAPI) for AI/ML microservices.
- **Database:** PostgreSQL (Amazon RDS) with PostGIS extension for geospatial queries.
- **Infrastructure:** AWS (EKS for container orchestration), Terraform for IaC.

PART 19: Roadmap

Phase	Timeline	Core Focus
MVP (Phase 1)	Months 0-6	Retail buyer diligence workflows, Partner App for Surveyors, basic Document parsing (English + 2 Regional).
Scale (Phase 2)	Months 7-18	Enterprise Portal launch, Advanced RAG for complex legal histories, Agentic workflow orchestration.
Vision (5 Year)	Years 2-5	The "Land Decision Graph" becomes the industry standard API for Banks and NBFCs to underwrite land loans instantly.

End of Document - BhumiSetu Confidential